

Advanced Modelling and Temporal Stability Analysis

NovaPay Fraud Detection — Experiment Report

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1. Introduction

This report documents the advanced modelling stage of the NovaPay fraud detection project. Building on the cleaned dataset and the baseline models established earlier, the work here had three aims: to train and tune gradient-boosting classifiers, namely XGBoost and LightGBM; to test several strategies for handling the severe class imbalance in the data; and to examine how stable model performance remains when the data is evaluated in chronological order rather than at random. The intent was not only to find the best-performing model, but to understand how trustworthy that performance is once the model is exposed to the way fraud actually arrives over time.

2. Methodology

The analysis used the cleaned NovaPay transaction dataset of roughly 11,200 records, in which fraudulent transactions make up about 8.8% of the total. Model performance is reported on a held-out test set of 2,240 transactions, of which 305 are fraudulent. All headline metrics describe performance on the fraud class, since that is the class of interest and the one most affected by imbalance.

Two gradient-boosting models were tuned across their main hyperparameters, including the number of estimators, learning rate, tree depth, and the positive-class weighting that compensates for imbalance. Their results were compared against the earlier Random Forest and Logistic Regression models to place them in context.

To address the imbalance directly, four approaches were compared under five-fold cross-validation: a baseline with no resampling, class-weight adjustment, random undersampling of the majority class, and SMOTE oversampling of the minority class. Because accuracy is misleading when one class dominates, the strategies were judged primarily on PR-AUC, the area under the precision-recall curve, which concentrates on performance on the rare fraud class.

Finally, to test stability, the data was divided into ordered time periods. The model was trained on earlier periods and evaluated on later ones, reproducing the conditions of a real deployment in which a model must predict transactions it has not yet seen. PR-AUC was recorded for each period.

3. Results

3.1 Model performance

The two boosting models performed strongly and very similarly to one another on the held-out test set. Both correctly identified 279 of the 305 fraudulent transactions, giving a recall of 0.91, and both missed the same 26 cases. The meaningful difference between them lay in the number of false alarms raised on legitimate transactions, where LightGBM was the more precise of the two.

Model	Precision	Recall	F1	ROC-AUC	False alarms
Logistic Regression	0.76	0.95	0.84	0.981	high
Random Forest	0.96	0.91	0.94	0.977	11
LightGBM	0.93	0.91	0.92	0.972	21
XGBoost	0.88	0.91	0.90	0.968	37

LightGBM achieved a precision of 0.93 and an F1 of 0.92, ahead of XGBoost at 0.88 and 0.90 respectively, owing to its lower false-alarm count of 21 against XGBoost's 37. Tuning narrowed the gap between the boosting models and the Random Forest, but did not close it. Random Forest retained the best precision and F1 with the fewest false alarms, and so remains the recommended production model. Logistic Regression recorded the highest recall at 0.95, but its low precision of 0.76 means it would flood analysts with false positives, making it unsuitable as the primary model.

3.2 Imbalance experiments

The four imbalance strategies produced results that were, in practical terms, indistinguishable. Under five-fold cross-validation, all four clustered around a PR-AUC of 0.835.

Strategy	PR-AUC	ROC-AUC
Baseline (no resampling)	0.836	0.923
Class-weight	0.835	0.920
Undersampling	0.835	0.921
SMOTE	0.837	0.925

It is worth noting that SMOTE executed correctly, balancing the training classes by oversampling the 690 fraud cases up to the 8,270 of the majority class, so the experiment is valid rather than a failed run. Despite this, none of the resampling approaches improved on the baseline by any meaningful margin. The conclusion drawn was to retain the simpler baseline. Resampling adds complexity and can distort the model's probability calibration, and there was no measurable performance gain to justify it.

3.3 Temporal stability

The chronological evaluation told a markedly different story from the random cross-validation. Where random splitting had produced a stable PR-AUC of around 0.84, the time-based evaluation showed performance ranging from near-zero in the earliest period to excellent in the latest.

Period (oldest to newest)	PR-AUC
1	0.033
2	0.490
3	0.918
4	0.919
5	0.954

Across the five periods the average PR-AUC was 0.66 with a standard deviation of 0.36, and the average ROC-AUC was 0.875 with a standard deviation of 0.122. The model was effectively unusable in the first period and highly effective by the last. This degree of variation, hidden entirely by random cross-validation, is the most important finding of the stage.

4. Reflection: the effect of varying fraud prevalence over time

The central insight from this stage is that the model's measured performance depends heavily both on how much fraud it has learned from and on how much fraud is present in the window being scored, and that both of these quantities change over time.

The collapse in the earliest period can be traced to the scarcity of fraud history. With very few fraudulent examples available to learn from, the model had not yet acquired the

patterns that distinguish fraud, and with few fraud cases in the evaluation window, the metric itself became volatile, so that a small number of errors moved precision sharply. As fraud history accumulated and patterns became better represented, performance rose steadily, reaching above 0.9 by the third period and remaining there.

A second effect is more subtle but equally important. PR-AUC is sensitive to the prevalence of the positive class, because the baseline score that a trivial classifier would achieve is approximately equal to the proportion of cases that are fraud. When fraud prevalence in a period is low, the achievable PR-AUC is naturally lower and noisier; when prevalence is higher, the same model can record a higher PR-AUC. Part of the swing across periods therefore reflects genuine model weakness from sparse training data, and part reflects a measurement artifact arising from shifting prevalence. Distinguishing the two is important before concluding how good the model truly is at any point in time.

Taken together, these effects show why a single random train-test split flatters a fraud model. Such a split implicitly assumes that fraud prevalence and fraud patterns are constant, which they are not. In practice, prevalence drifts with seasonality, with coordinated attack campaigns, with the introduction of new products or corridors, and with the effectiveness of existing defences. As prevalence and tactics move, a fixed model and a fixed decision threshold will see their recall and precision fluctuate, and the observed range of 0.03 to 0.95 is a direct demonstration of how large that fluctuation can be.

The practical implications follow naturally. Performance should be evaluated on a rolling, time-forward basis and tracked period by period, not summarised in a single figure. Fraud prevalence should be monitored as a signal in its own right, since a change in the base rate is both a business event and a reason to expect metric movement and possibly to recalibrate the decision threshold. And the model should be retrained on a schedule as fraud history accumulates, because a model that performs well today can decay as patterns shift. These conclusions are the basis for the project's recommendation that the model be deployed alongside drift monitoring, threshold review, and scheduled retraining, rather than as a single deployment assumed to remain valid indefinitely.

5. Conclusion

The advanced modelling stage confirmed that gradient-boosting models, particularly LightGBM, are competitive but do not surpass the Random Forest, which remains the recommended production model. It established that the class imbalance does not require resampling, since none of the four tested strategies improved on a simple baseline. Most importantly, the temporal analysis revealed that performance is far less stable over time than random validation implies, and that fraud prevalence is a moving quantity which drives both the model's learning and the metrics used to judge it. Treating prevalence as constant is the most direct way to overstate a fraud model's reliability, and the recommendations of this project are designed to guard against exactly that.